

TP Modul 04

Nama : William Peter Vanx Najohan

NIM : 103032400084

Kelas : IT-48-01

Kode Asprak : MSQ

Soal 1

main.cpp

```
1 | #include "list.h"
2 |
3 | using namespace std;
4 |
5 | int main(){
6 |     int angka1, angka2, angka3;
7 |     List L;
8 |     address ak1, ak2, ak3;
9 |
10 |    createList_103032400084(L);
11 |
12 |    cout << "Angka pertama untuk di input: " << endl;
13 |    cin >> angka1;
14 |
15 |    ak1 = allocate_103032400084(angka1);
16 |    insertFirst_103032400084(L, ak1);
17 |
18 |    printInfo_103032400084(L);
19 |
20 |    cout << "Angka pertama untuk di input: " << endl;
21 |    cin >> angka2;
22 |
23 |    ak2 = allocate_103032400084(angka2);
24 |    insertFirst_103032400084(L, ak2);
25 |
26 |    printInfo_103032400084(L);
27 |
28 |    cout << "Angka pertama untuk di input: " << endl;
29 |    cin >> angka3;
30 |
31 |    ak3 = allocate_103032400084(angka3);
32 |    insertFirst_103032400084(L, ak3);
33 |
34 |    printInfo_103032400084(L);
35 |
36 |    return 0;
37 | }
38 |
```

list.cpp

```
1  #include <iostream>
2  #include "list.h"
3
4  using namespace std;
5
6  void createList_103032400084(List &L) {
7      L.first = nullptr;
8  }
9
10 address allocate_103032400084(intofype x) {
11     address p = new elmlist;
12
13     p -> info = x;
14     p -> next = nullptr;
15
16     return p;
17 }
18
19 void insertFirst_103032400084(List &L, address p) {
20     p -> next = L.first;
21     L.first = p;
22 }
23
24 void printInfo_103032400084(List L) {
25     address p = L.first;
26
27     while (p) {
28         cout << p -> info << ", ";
29         p = p -> next;
30     }
31
32     cout << endl;
33 }
```

list.h

```
1  #ifndef LIST_H_INCLUDED
2  #define LIST_H_INCLUDED
3
4  #include <iostream>
5
6  using namespace std;
7
8  typedef int infotype;
9  typedef struct elmlist *address;
10
11  struct elmlist {
12      infotype info;
13      address next;
14  };
15
16  struct List {
17      address first;
18  };
19
20  address allocate_103032400084(infotype x);
21
22  void createList_103032400084(List &L);
23  void insertFirst_103032400084(List &L, address p);
24  void printInfo_103032400084(List L);
25
26  #endif // LIST_H_INCLUDED
```

output

```
Angka pertama untuk di input:
0
0,
Angka pertama untuk di input:
8
8, 0,
Angka pertama untuk di input:
4
4, 8, 0,

Process returned 0 (0x0)   execution time : 19.614 s
Press any key to continue.
```